Magnetism in Rocks

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Aim:

Calculate the magnetic field strength of different rocks/ minerals using a smart device magnetometer app. Understand how remanent magnetism in crustal rocks contributes to Earth's total magnetic field, and how geomagnetic surveying is used to identify magnetic anomalies.

Materials:

- Smart device with the phyphox app downloaded & access to the internet
- Steel nail
- Selection of different minerals/ rocks no larger than hand-sized (ideally a sample of magnetite, an igneous rock and a sedimentary rock)

Activity 1: Predicting the magnetic properties of your samples

Using your existing understanding of magnetism and rocks, order your rock/ mineral samples from most to least magnetic.

1. Which sample do you think is the <u>most</u> magnetic? Why?

2. Which sample do you think is the <u>least</u> magnetic? Why?

Activity 2: Measuring the magnetic field strength of your samples using the phyphox app

We will now quantify the magnetic field strength of your samples using a magnetometer app on your smart device.

Note: You will need to have the phyphox app downloaded on your device for this activity. See the document *'Downloading and using the phyphox magnetometer'* for guidance.

<u>Step 1: Locating the Magnetic Sensor in your Smart Device</u>

Demonstration video: Magnetism in Rocks Step 1: Locating the Magnetic Sensor in your Smart Device

Before we begin collecting geomagnetic data, it is important to know where the magnetometer sensor in your smart device is located.

You have been given a steel nail.

1. Place your device on a flat surface. Do not move it during the experiment.
2. Begin recording data with the phyphox magnetometer in **Graph** mode. Allow the readings to stabilise for 5-10 seconds.
3. Slowly move the nail across your device screen from bottom to top (be careful not to scratch your screen!).
4. The readings on all 3 graphs will spike when the nail is close to the sensor (see **Figure 1**).

In newer iPhone devices (iPhone X and onwards), this should be the top left corner.

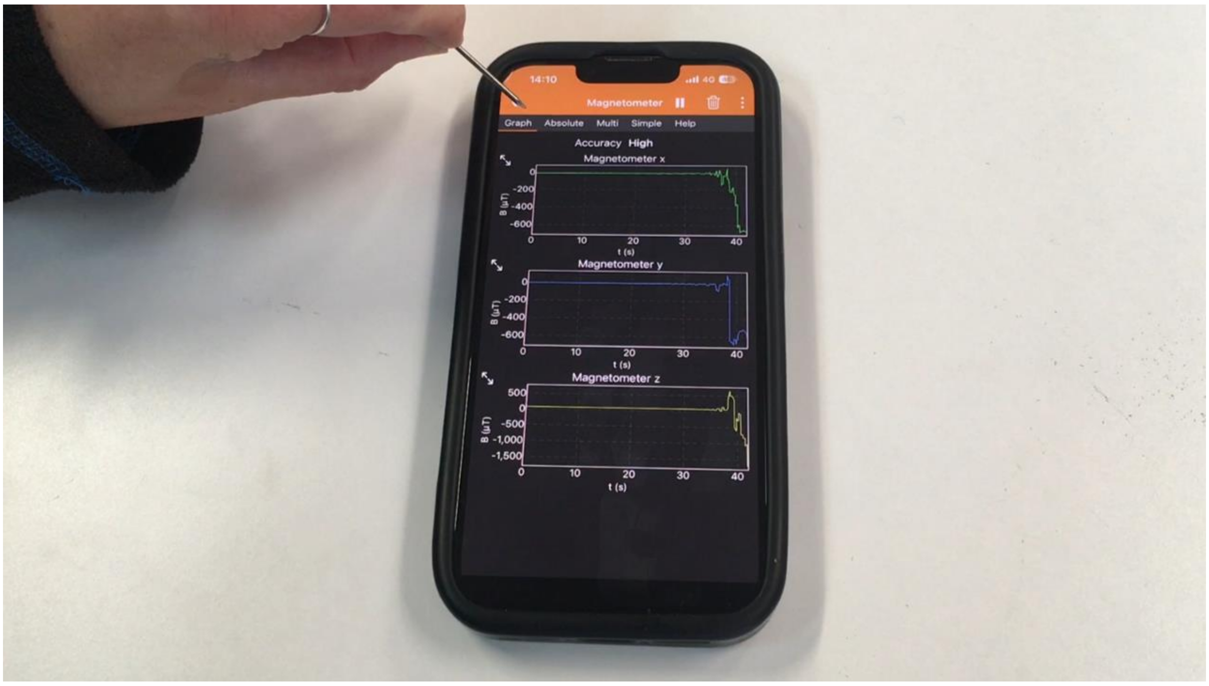

Figure 1: phyphox magnetometer readings spike when the nail is close to the phone's magnetometer sensor (top left corner in this iPhone 13).

<u>Step 2: Measuring the Magnetic Field Strength of Rocks using Phyphox</u>

Demonstration video: [Magnetism in Rocks Step 2: Measuring the Magnetic Field Strength of Rocks using Phyphox]

1. Place the device on a table away from metal/ electronic objects. <u>DO NOT MOVE</u> the phone from this position during the experiment.

2. Begin recording using the **Simple** magnetometer mode in phyphox. Allow the readings to stabilise for 5-10 seconds.
3. Then record the **Absolute** magnetic value – this is **Earth's ambient magnetic field (A).**

Note: the absolute value is the total magnetic field and is calculated from the x, y and z components using the Pythagorean theorem $B_{Absolute} = \sqrt{B_x^2 + B_y^2 + B_z^2}$

4. Leave your phone in its current position. Now place your rock/ mineral sample close to/ over the magnetic sensor in your device.
5. Allow the new readings to stabilise for 10 seconds then record the new Absolute magnetic value – this is the **total magnetic field (T).**

Note: the readings may change rapidly if your sample has a strong magnetic field. Hold the sample in one position over the magnetometer sensor and record the average value as best you can.

6. Now we will isolate the **magnetic field of your sample (S)** by subtracting the value of Earth's ambient magnetic field (A) from the total magnetic field (T). An example is provided below in **Table 1**.

Table 1: Example magnetic measurements for a rock sample.

Rock/Sample name	Ambient magnetic field (A) /μT	Total magnetic field (T) /μT	Sample magnetic field (S) (= T – A) /μT
Example	64	72	72 – 64 = 8

If the value of S is greater than ±0.5 μT, phyphox has detected that your sample has a magnetic field.

Repeat this process for each of your samples.

Recalibrate the device's magnetometer before measuring each sample to reset the readings. To do this, move it in a sweeping figure-8 motion as shown in the demonstration video.

The samples that are the most magnetic will have a higher total field strength.

Rank your samples again from most to least magnetic based on the total field strengths you calculated. Was your prediction in **Activity 1** correct?

Activity 3: An introduction to magnetic anomalies and geomagnetic surveying

Navigate to the Geoscience Australia Portal: https://portal.ga.gov.au/

What is Geoscience Australia?

Geoscience Australia (GA) is a government agency that conducts geoscientific research. They advise the government on everything related to geoscience, and store geological and geographical data collected by the commonwealth (Wikipedia, 2026).

The portal opens a map of Australia.

Select the 'Layers' option from the toolbox and in the search bar type 'Total Magnetic Intensity'.

Select **'Magnetic Intensity Image'** from the list and click 'Add to Map'.

Your map should look like the following image (**Figure 2**):

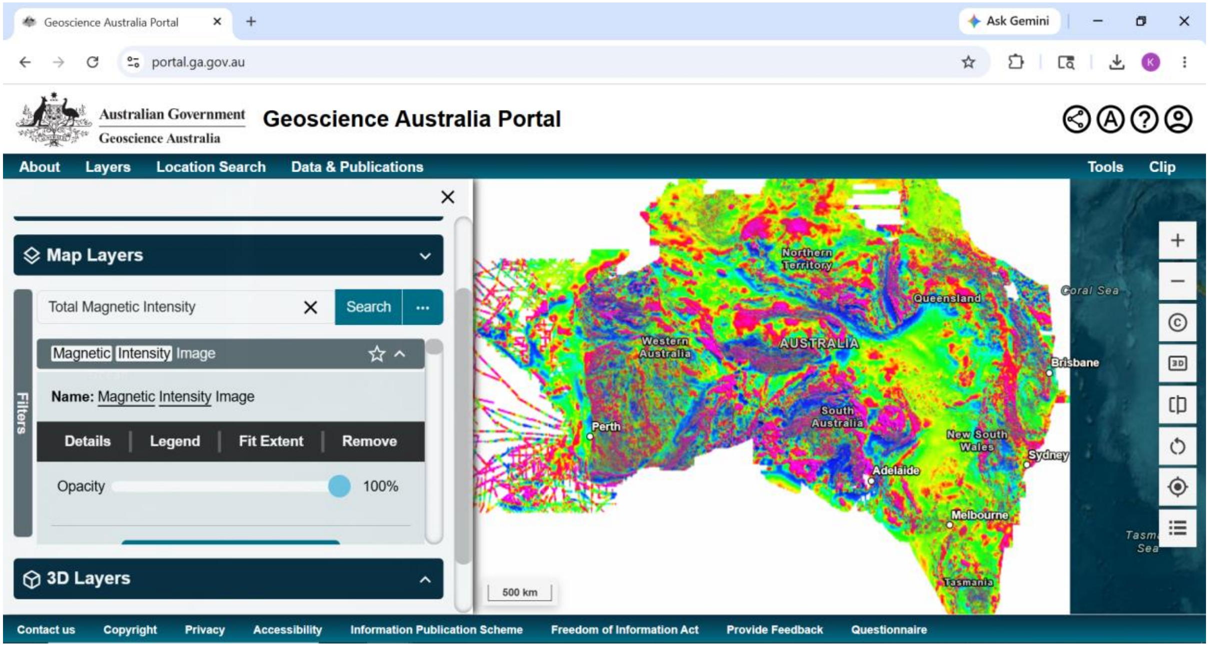

Figure 2: Total Magnetic Intensity (TMI) image of Australia (Minty and Poudjom Djomani, 2019). Viewed via Geoscience Australia Portal (https://portal.ga.gov.au/).

Information about this dataset can be found here.

The map shows the magnetic field strength of crustal rocks. This was measured by magnetometers during geomagnetic surveying. The scale is in nanotesla (nT) (1 nT = 0.001 μT).

Magnetic anomalies are variations in the geomagnetic field due to the differing magnetic properties of rocks in the lithosphere (Luyendyk, 2025).

- Strong magnetic anomalies are coloured bright red/ pink or a deep blue. These are areas of strongly magnetic rocks.

- Weak magnetic anomalies are coloured green or yellow. These are areas of weak or non-magnetic rocks.

Explore the map using the zoom and pan controls.

- Zoom using the + or - buttons on the right of the screen, the scroll wheel on a mouse, or 'pinch to zoom' if your device is touchscreen.
- Pan by clicking and dragging the screen or swiping across the screen on a touchscreen device.

Based on your study of the map and your experimentation with rock samples in **Activity 2**, answer the following questions:

1. What might strong magnetic anomalies indicate about the geology of an area?

2. What might weak magnetic anomalies indicate about the geology of an area?

HINT: Navigate to Hobart and zoom in until you can see the bright red/ pink rounded shapes to the west of the city – this is the Wellington Range. The Wellington Range is primarily composed of **dolerite** – an igneous rock rich in the magnetic mineral magnetite. These dolerites are strongly magnetic, so appear as a strong magnetic anomaly on the map.

References

Luyendyk, B. P. (2025). Oceanic crust. *Encyclopedia Britannica*. Available at: https://www.britannica.com/science/oceanic-crust

Minty, B.R.S. and Poudjom Djomani, Y. (2019). Total Magnetic Intensity (TMI) grid of Australia 2019: seventh edition, 40 m cell size. Geoscience Australia, Canberra. Available at: https://doi.org/10.26186/5dd5fb557fffb (Accessed: 14 May 2026). Viewed via Geoscience Australia Portal.

Wikipedia (2026). *Geoscience Australia, Wikipedia*. Available at: https://en.wikipedia.org/wiki/Geoscience_Australia (Accessed: 15 June 2026).

This resource was developed with assistance from ChatGPT (OpenAI, 2026), used to generate and refine instructional ideas.